

Course Project Part 3 ReadMe

1. Commits:

a. Refactor 1: Decompose compositional

This commit creates a new function called `edgeHasSrcAndDst`. This function allows for the `removeEdge` function to be easier to understand because the conditional expression has been abstracted.

<https://github.com/HKosten/CSE464CourseProject1/commit/7500a7c0e3dd96391ecf431e821b61e134c74a1f>

b. Refactor 2: Change function declaration

This changes the name of the `addNodes` and `removeNodes` functions to `addListOfNodes` and `removeListOfNodes`. This change makes it easier to understand what the function does just based on the signature.

<https://github.com/HKosten/CSE464CourseProject1/commit/8642a12ecf3e85edeb5e58c373737bc7cd76279b>

c. Refactor 3: Method Extract

This creates a new method called `setUpImporter`. This method abstracts the setup functions that need to be invoked on the importer. This makes it easier to understand the flow of `parseGraph`.

<https://github.com/HKosten/CSE464CourseProject1/commit/e48e5d844f78cd9c3a6ff15a07e8a46a7a299587>

d. Refactor 4: Inline variable

This removes the need for the constructor method. The original constructor did not have any parameters and therefore the initialization of the constructor could be done inline with the declaration of the `Graph`.

<https://github.com/HKosten/CSE464CourseProject1/commit/2896894185c7ea931c08847a253880538a91f67c>

e. Refactor 5: Extract method

This change creates a new method called `executeOutput`. `executeOutput` is intended to make the `outputGraph` method easier to read. It does this by extracting the confusing code that is needed to execute the output.

<https://github.com/HKosten/CSE464CourseProject1/commit/bd7cada5c122583ded39f6e736f788a5a4df00f2>

2. Commits:

a. Refactor Template Model:

This change created 2 new classes BFS and DFS. These classes extend MyGraph. This allows for the GraphSearch method to invoke a new abstract method called iteratorAlg. iteratorAlg is overridden by the BFS and DFS classes.

<https://github.com/HKosten/CSE464CourseProject1/commit/aff3a9067d78104d46899d79f465d5b3647628ba>

3. Commits:

a. Refactor Strategy Model:

This change implemented a new interface SearchStrategy (instead of the MyGraph class). The BFS and DFS classes now implement the SearchStrategy interface and still override the iteratorAlg method. The MyGraph class now has an attribute called ss. ss stores a SearchStrategy object. There is a new method called setSearchStrategy. setSearchStrategy sets the ss attribute to a DFS object or a BFS object based upon the algo parameter.

<https://github.com/HKosten/CSE464CourseProject1/commit/2273bd9d5d12f68235208d287348bf600f5e7cfa>

b. Refactor Strategy + Template Pattern:

SearchStrategy is now an abstract class instead of an interface. its iteratorAlg is now abstract as well. BFS and DFS now extend SearchStrategy.

<https://github.com/HKosten/CSE464CourseProject1/commit/ae195373f179c25cb07e750e52e111d3d14e10c1>

4. Commits:

a. Refactor RWS:

This change added an RWS class that extends the SearchStrategy class. The RWS class can be set as the MyGraphs search strategy. The RWS class overrides the iteratorAlg so that it returns a RandomWalkIterator<String> object. Also added a Main class that parses the input.dot and searches it with the RWS strategy.

<https://github.com/HKosten/CSE464CourseProject1/commit/263e8f21c01312aee32031b36a9140ad1d434ae4>

b. Refactor RWS:

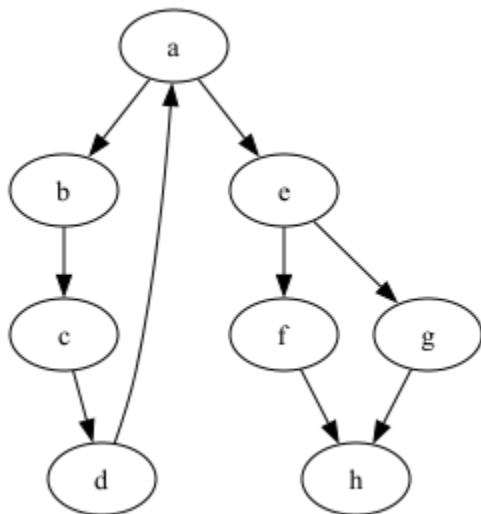
This change makes the tracePath method belong to the SearchStrategy class. Its original implementation has not been changed; However, it is overridden in the RWS class in order to make the path reset when it finds a loop or a dead end. (This change was suggested by the TA during my demo.)

<https://github.com/HKosten/CSE464CourseProject1/commit/26d4c30ed730dd148a83ca20bb1316b899536226>

The code in the Main.java file (image below) parses the graph in the canvas files that looks like this:

```
import java.io.IOException;

public class Main {
    public static void main(String[] args) throws IOException {
        MyGraph myg = new MyGraph();
        myg.parseGraph(filepath: "input.dot");
        myg.graphSearch(src: "a", dst: "c", Algorithm.RWS);
    }
}
```



The Main.java file then runs the graphSearch() function with parameters that tell it to do a random walk from node “a” to node “c”.

This function causes the following outputs:

```
random testing
visiting Path{nodes=[Node{a}]} visiting Path{nodes=[Node{a}, Node{b}]} visiting Path{nodes=[Node{a}, Node{b}, Node{c}]}
Process finished with exit code 0
```

```
random testing
visiting Path{nodes=[Node{a}]} visiting Path{nodes=[Node{a}, Node{e}]} visiting Path{nodes=[Node{a},
Node{e}, Node{f}]} visiting Path{nodes=[Node{a}, Node{e}, Node{f}, Node{h}]} visiting
Path{nodes=[Node{a}]} visiting Path{nodes=[Node{a}, Node{b}]} visiting Path{nodes=[Node{a}, Node{b},
Node{c}]}
Process finished with exit code 0
```

```
random testing
visiting Path{nodes=[Node{a}]} visiting Path{nodes=[Node{a}, Node{e}]} visiting Path{nodes=[Node{a},
Node{e}, Node{f}]} visiting Path{nodes=[Node{a}, Node{e}, Node{f}, Node{h}]} visiting
Path{nodes=[Node{a}]} visiting Path{nodes=[Node{a}, Node{e}]} visiting Path{nodes=[Node{a}, Node{e},
Node{f}]} visiting Path{nodes=[Node{a}, Node{e}, Node{f}, Node{h}]} visiting Path{nodes=[Node{a}]}
visiting Path{nodes=[Node{a}, Node{b}]} visiting Path{nodes=[Node{a}, Node{b}, Node{c}]}
Process finished with exit code 0
```

You can see that in each output the random walk goes down the wrong path a different number of times before finding the path to “c”. The first image goes wrong 0 times. The second goes wrong 1 time. The third goes wrong 2 times.

5. Pull link:

<https://github.com/HKosten/CSE464CourseProject1/pull/1>

Instructions:

1. Initialize a MyGraph object like this:
 - a. `MyGraph myg = new MyGraph();`
2. Parse or build the graph like this:
 - a. `myg.parseGraph(String filepath);`
 - b. `myg.addNodes(String[] nodes); myg.addVertex(String src, String dst);`
3. Search the graph like this:
 - a. `myg.graphSearch(String src, String dst, Algorithm algo);`
 - i. Note: for the 3rd parameter your choices are `Algorithm.BFS`, `Algorithm.DFS`, `Algorithm.RWS`

Example code:

```
import java.io.IOException;

public class Main {
    public static void main(String[] args) throws IOException {
        MyGraph myg = new MyGraph();
        myg.parseGraph(filepath: "input.dot");
        myg.graphSearch(src: "a", dst: "c", Algorithm.RWS);
    }
}
```

Contiguous integration: The green check marks under each commit show that all CI tests have passed.

The screenshot displays a commit history interface with three sections, each showing a commit with a green checkmark indicating successful CI tests.

- Commits on May 5, 2026**
 - Merge pull request #1 from HKosten/refactor** (Verified, 8801c94)
 - HKosten authored 11 minutes ago · 1 / 1
 - Refactor RWS: This change makes the tracePath method belong to the SearchStrategy class. Its original implementation has not been changed. However, it is overridden in the RWS class in order to mak...** (26d4c30)
 - HKosten committed 5 hours ago · 1 / 1
- Commits on May 1, 2026**
 - Refactor RWS: This change added an RWS class that extends the SearchStrategy class. The RWS class can be set as the MyGraphs search strategy. The RWS class overrides the iteratorAlg so that it retu...** (263e8f2)
 - HKosten committed 4 days ago · 1 / 1
 - Refactor Strategy + Template Pattern: SearchStrategy is now an abstract class instead of an interface. its iteratorAlg is now abstract as well. BFS and DFS now extend SearchStrategy.** (ae19537)
 - HKosten committed 4 days ago · 1 / 1
- Commits on Apr 30, 2026**
 - Refactor Strategy Model: This change implemented a new interface called SearchStrategy. The SearchStrategy interface is now the owner of the iteratorAlg method (instead of the MyGraph class). The B...** (2273bd9)
 - HKosten committed 5 days ago · 1 / 1
 - Refactor Template Model: This change created 2 new classes BFS and DFS. These classes extend MyGraph. This allows for the GraphSearch method to invoke a new abstract method called iteratorAlg. iter...** (aff3a90)
 - HKosten committed 5 days ago · 1 / 1